HARMONIC SERIES

Terms to zero, sum to what?

Terms approaching zero is necessary for convergence but not sufficient.

CONVERGENCE

$$\sum a_n \text{ converges}$$

HOLDS WHEN the terms must approach 0, and must shrink fast enough

✓ Converges

$$\sum \frac{1}{2^n} = 1$$

Geometric terms shrink by a constant factor: fast enough to keep the total finite.

BOUNDARY

✗ Diverges

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

The terms shrink to zero, but too slowly: each doubling block of terms adds at least $\frac{1}{2}$, and there are infinitely many blocks.

HOW SLOWLY $\ln n + \gamma$ The partial sums approximate $\ln n + \gamma$ with $\gamma \approx 0.5772$; passing 10 takes over $12{,}000$ terms.